Q1. Convert RGB (30,108,210) to HSI and CMY

$R = 30,\ G = 108,\ B = 210$

Intensity:

$I = (R + G + B) / 3 = (30 + 108 + 210) / 3 = 116$

Saturation:

$S = 1 - (3 \times \min(R,G,B)) / (R + G + B)$

$S = 1 - (3 \times 30)/348 = 0.74$

Hue:

$\theta = \arccos\{ 0.5[(R-G)+(R-B)] / \sqrt{[(R-G)^2 + (R-B)(G-B)]} \}$

$\theta \approx 145.4°$

Since $B > G \longrightarrow H = 360° - \theta = 214.6°$

$HSI = (H = 214.6°,\ S = 0.74,\ I = 116)$

CMY

$C = 1 - R/255 = 0.882$

$M = 1 - G/255 = 0.576$

CMY

$C = 1 - R/255 = 0.882$

$M = 1 - G/255 = 0.576$

$Y = 1 - B/255 = 0.176$

$CMY \approx (0.882, 0.576, 0.176)$

3D Representation:

RGB → point (30,103,210) in RGB cub

HSI → (H angle, S radius, I height) in cylinder

CMY → point (0.882,0.576,0.176) in CMY cube

Q2. Intensity slicing for X-ray

Choose thresholds T1, T2 from histogram.

0–T1 → black (background)

T1–T2 → highlighted slice (soft tissue)

>T2 → white (bone)

Use narrow slices for soft tissue to make contrasts visible.

Q3. Pseudo-color after smoothing

Smoothing removes noise → reveals true intensity differences.
Pseudo-color maps small grayscale differences to strong color changes.
Hidden edges/patterns appear clearly after color mapping.

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Pseudo-color for medical grayscale data

Intensity ∝ tissue density.
Fat = low, Muscle = mid, Bone = high

Map intensities to colors.
Fat → blue/cyan
Muscle → green/yellow
Bone → orange/red

Radiologist can easily distinguish tissues using color-coded densities.

Q5. False color composite in satellite images

Assign NIR → R, Red → G, Green → B
Vegetation appears bright red
Water becomes dark
Urban areas become distinct
Helps land cover classification, crop monitoring, water detection.

Q6. Uncompressed video size

Resolution: 1024 × 720
Pixels/frame = 780,480
Bytes/frame = 780,480 × 3 = 2,341,440

Total frames = 80 × 10,800 = 864,000
Total size = 2,341,440 × 864,000 = 2.023 × 10^{12} bytes
≈ 2.023 TB

(b) Blu-ray discs needed:
2.023 TB / 50 GB ≈ 41 discs

(b) Blu-ray discs needed:
2.023 TB / 50 GB $\approx$ 4 discs

(c) Transmission time @100 Mbps:
Total bits $= 2.023 \times 10^{12} \times 8 = 1.618 \times 10^{13}$
Time = bits / 100×10^6 = 161,840 s
= 44.96 hours $\approx$ 1.87 days

Q7. Huffman coding

Symbol frequencies (out of 25):
0:1, 1:4, 2:5, 3:2, 5:2, 6:3, 7:2, 8:3, 9:3

One Huffman code:
0 $\longrightarrow$ 1010
1 $\longrightarrow$ 110
2 $\longrightarrow$ 00
3 $\longrightarrow$ 1011
5 $\longrightarrow$ 1110
6 $\longrightarrow$ 010
7 $\longrightarrow$ 1111
8 $\longrightarrow$ 011
9 $\longrightarrow$ 100

Average code length $L \approx 3.08$ bits

Average code length $L \approx 3.08$ bits
Entropy $H \approx 3.049$ bits
Total bits $= 25 \times 3.08 = 77$ bits
Original (8-bit pixels) $= 200$ bits
Compression ratio $= 200 / 77 \approx 2.6 : 1$

Q8. Why DSP preferred over FPGA/ASIC

Built-in MAC units, shifters, circular buffers
Deterministic real-time execution
Lower power and cost
Easier programming and faster development
Ideal for audio, motor control, filters

Q9. Five-stage DSP pipeline

Stages:

1. IF — fetch
2. ID — decode

Q2. Five-stage DSP pipeline

Stages:

1. IF — fetch
2. ID — decode
3. AGU — address generation
4. EX — execute (MAC/ALU)
5. WB — write back

Pipeline flow:
IF | ID | AGU | EX | WB (parallel for consecutive instructions)

Hazards:
Data hazard → stall
Control hazard → flush
Structural hazard → resource conflict

Q10. Why DSPs for real-time processing

Key features:
Single-cycle MAC → fast filtering
Harvard architecture → fetch + data access same cycle
Circular buffering → zero overhead loops
Deep pipelining / parallel ALUs → high throughput Deterministic interrupt response

Comparison:
ASIC → fastest but inflexible
FPGA → parallel but high power and complex
GPU → high latency, non-deterministic

DSP → best real-time, low-latency, programmable solution.